Common setup (run once)

```
pip install nltk
```

```
import nltk

# Download the NLTK datasets/models needed by the lab programs.
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('brown')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('wordnet')
nltk.download('omw-1.4')
nltk.download('universal_tagset')
nltk.download('conll2000')
nltk.download('maxent_ne_chunker_tab')
nltk.download('words')
```

Run each program in Jupyter Notebook, Google Colab, or VS Code with Python installed. Some programs download corpora on first run and need an internet connection.

1. Word counting, vocabulary operations, and list processing

(i) Count the total number of words and distinct words

```
# Store a sample text document in a string.
text = """
She sells sea shells by the sea shore.
She sells beautiful shells.
"""

# Convert the complete text to lowercase so that words like
# "She" and "she" are treated as the same word.
text = text.lower()

# Split the text into individual words.
words = text.split()

# Count the total number of words, including repetitions.
total_words = len(words)

# Convert the word list into a set to remove duplicates.
distinct_words = set(words)

# Print the results.
print("Total number of words:", total_words)
print("Number of distinct words:", len(distinct_words))
print("Distinct words:", sorted(distinct_words))
```

Sample output

```
Total number of words: 12
Number of distinct words: 8
Distinct words: ['beautiful', 'by', 'sea', 'sells', 'shells.', 'she', 'shore.', 'the']
```

Explanation: `split()` separates the text into words, `len()` counts them, and `set()` removes duplicate entries. This basic example retains punctuation attached to words.

(ii) Compute the vocabulary of sentences sent1 to sent8

```
from nltk.book import sent1, sent2, sent3, sent4
from nltk.book import sent5, sent6, sent7, sent8

# Combine all eight sentences using list addition.
sentences = sent1 + sent2 + sent3 + sent4 + sent5 + sent6 + sent7 + sent8

# Convert the combined list into a set to get unique tokens.
vocabulary = set(sentences)

# Sort the vocabulary alphabetically.
sorted_vocabulary = sorted(vocabulary)
```

```
print("Total tokens:", len(sentences))
print("Vocabulary size:", len(vocabulary))
print("First 20 vocabulary items:", sorted_vocabulary[:20])
```

Sample output

```
Total tokens: [depends on the NLTK book sentences]
Vocabulary size: [depends on the NLTK book sentences]
First 20 vocabulary items: [alphabetically sorted words]
```

The exact numbers are obtained when the program runs with the NLTK book corpus. `sent1` through `sent8` are example sentences defined in NLTK's `book` module.

(iii) Perform operations on the given list

```
# Define the given list of words.
sent = ['she', 'sells', 'sea', 'shells', 'by',
        'the', 'sea', 'shore']

# (a) Print words beginning with "sh".
print("Words beginning with 'sh':")

for word in sent:
    if word.startswith("sh"):
        print(word)

# (b) Print words longer than four characters.
print("Words longer than four characters:")

for word in sent:
    if len(word) > 4:
        print(word)
```

Sample output

```
Words beginning with 'sh':
she
shells
shore

Words longer than four characters:
sells
shells
shore
```

Explanation:

• `startswith("sh")` checks the beginning of each word.

• `len(word) > 4` selects words containing more than four characters.